MILITARY SPECIFICATION SHEET

ADAPTERS, COAXIAL TO WAVEGUIDE
(SERIES SMA TO WAVEGUIDE - CLASS 5)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-A-22641.

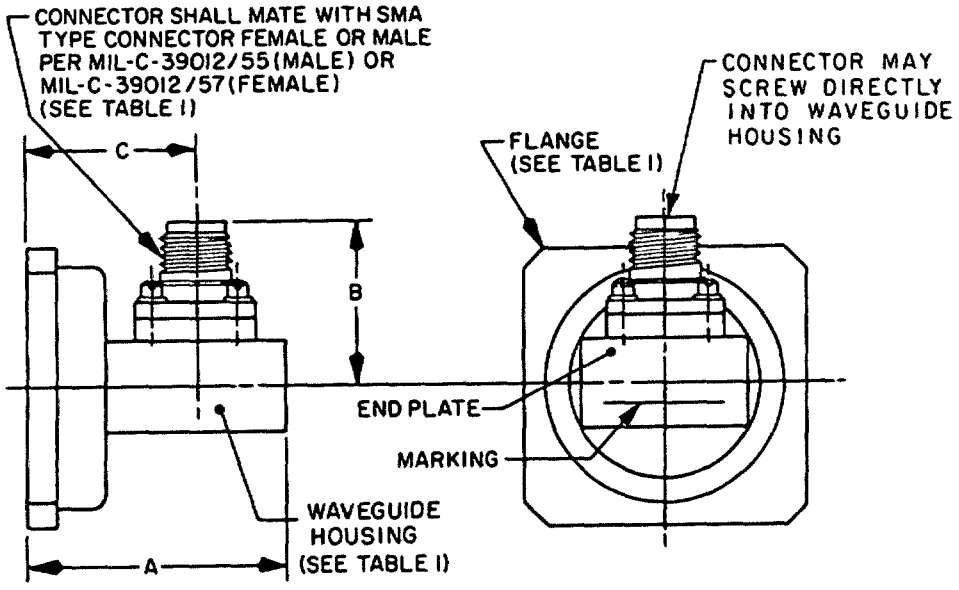


FIGURE 1. Adapters, coaxial to waveguide (class 5).

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.

Operating temperature range: -54° to +125°C.

Frequency range: See table I.

VSWR: See table I.

RF insertion loss: See table I.

Connector: Connector bodies shall be corrosion-resistant steel.

Flange, waveguide: See table I.

Housing, waveguide: See table I.

Marking: See figure 1.

Part number: M22641/20- (dash number from table I).

FSC 5985

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TABLE I. Adapters, coaxial to waveguide (class 5 - SMA to waveguide).

Dash number	Frequency range (GHz)	VSWR over entire frequency range (max) 4/ 5/	VSWR over 50% frequency range (max) 4/ 5/	RF insertion loss dB (max) 5/	Waveguide housing 6/	Flange 7/	Waveguide and flange material	Connector	Dimensions 1/ 2/ 3/ A (max)	B (max)	C (max)
01	2.60 - 3.95	1.10:1	1.05:1	0.05	M85/1-041 (RG-75/U)	M3922/56-002 (UG-584/U)	Aluminum	Male	2.235 (56.77)	1.219 (30.96)	1.269 (32.23)
02	2.60 - 3.95	1.10:1	1.05:1	0.05	M85/1-041 (RG-75/U)	M3922/56-002 (UG-584/U)	Aluminum	Female	2.235 (56.77)	1.219 (30.96)	1.349 (34.26)
03	8.2 - 12.4	1.10:1	1.05:1	0.15	M85/1-077 (RG-67/U)	M3922/53-003 (UG-135/U)	Aluminum	Male	1.296 (32.92)	0.807 (20.50)	0.940 (23.88)
04	8.2 - 12.4	1.10:1	1.05:1	0.15	M85/1-077 (RG-67/U)	M3922/53-003 (UG-135/U)	Aluminum	Female	1.296 (32.92)	0.734 (18.64)	0.940 (23.88)
05	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-090 (RG-349/U)	M3922/53-006 (UG-1665/U)	Aluminum	Male	1.265 (32.13)	0.782 (19.86)	0.941 (23.90)
06	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-090 (RG-349/U)	M3922/53-006 (UG-1665/U)	Aluminum	Female	1.265 (32.13)	0.687 (17.45)	0.941 (23.90)
07	2.60 - 3.95	1.10:1	1.05:1	0.05	M85/1-043 (RG-48/U)	M3922/56-001 (UG-53/U)	Copper	Male	2.235 (56.77)	1.219 (30.96)	1.269 (32.23)
08	2.60 - 3.95	1.10:1	1.05:1	0.05	M85/1-043 (RG-48/U)	M3922/56-001 (UG-53/U)	Copper	Female	2.235 (56.77)	1.219 (30.96)	1.269 (32.23)
09	8.2 - 12.4	1.10:1	1.05:1	0.15	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)	Copper	Male	1.296 (32.92)	0.807 (20.50)	0.940 (23.88)
10	8.2 - 12.4	1.10:1	1.05:1	0.15	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)	Copper	Female	1.296 (32.92)	0.807 (20.50)	0.940 (23.88)
11	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)	Copper	Male	1.256 (31.90)	0.734 (18.64)	0.940 (23.88)
12	12.4 - 18.0	1.25:1	1.05:1	0.20	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)	Copper	Female	1.256 (31.90)	0.782 (19.86)	0.940 (23.88)
13	7.0 - 11.0	1.10:1	1.05:1	0.15	M85/1-158	M3922/70-014	Aluminum	Male	1.350 (34.29)	0.970 (24.64)	0.950 (24.13)
14	7.0 - 11.0	1.10:1	1.05:1	0.15	M85/1-158	M3922/70-014	Aluminum	Female	1.350 (34.29)	0.970 (24.64)	0.950 (24.13)
15	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-090 (RG-349/U)	M3922/70-020	Aluminum	Male	1.265 (32.13)	0.910 (23.11)	0.840 (21.34)
16	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-090 (RG-349/U)	M3922/70-020	Aluminum	Female	1.265 (32.13)	0.910 (23.11)	0.840 (21.34)
17	7.0 - 11.0	1.10:1	1.05:1	0.15	M85/1-155 (RG-320/U)	M3922/70-013	Copper	Male	1.350 (34.29)	0.970 (24.64)	0.950 (24.13)
18	7.0 - 11.0	1.10:1	1.05:1	0.15	M85/1-155 (RG-320/U)	M3922/70-013	Copper	Female	1.350 (34.29)	0.970 (24.64)	0.950 (24.13)
19	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-089 (RG-91/U)	M3922/70-019	Copper	Male	1.265 (32.13)	0.910 (23.11)	0.840 (21.34)
20	12.4 - 18.0	1.10:1	1.05:1	0.20	M85/1-089 (RG-91/U)	M3922/70-019	Copper	Female	1.265 (32.13)	0.910 (23.11)	0.840 (21.34)
21 8/	2.60 - 3.95	1.04:1	---	0.05	M85/4-017	M3922/75-06	Aluminum	Male	2.50 (63.5)	.913 (23.19)	1.53 (38.9)
22 8/	2.60 - 3.95	1.04:1	---	0.05	M85/4-017	M3922/75-06	Aluminum	Female	2.50 (63.5)	.913 (23.19)	1.53 (38.9)

See footnotes at end of table.

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TABLE I. Adapters, coaxial to waveguide (class 5 - SMA to waveguide) -Continued.

Dash number	Frequency range (GHz)	VSWR over entire frequency range (max) 4/5/	VSWR over 50% frequency range (max) 4/5/	RF insertion loss dB (max) 5/	Waveguide housing 6/	Flange 7/	Waveguide and flange material	Connector	Dimensions 1/2/3/		
									A (max)	B (max)	C (max)
23 8/	2.60 - 5.85	1.25:1	---	0.05	M85/ 4-007	M3922/ 75-02	Aluminum	Male	2.50 (63.5)	1.10 (27.9)	1.60 (40.6)
24 8/	2.60 - 5.85	1.25:1	---	0.05	M85/ 4-007	M3922/ 75-02	Aluminum	Female	2.50 (63.5)	1.10 (27.9)	1.60 (40.6)
25 8/	5.85 - 8.20	1.04:1	---	0.05	M85/ 4-008	M3922/ 75-04	Aluminum	Male	1.50 (38.1)	.88 (22.4)	1.13 (28.7)
26 8/	5.85 - 8.20	1.04:1	---	0.05	M85/ 4-008	M3922/ 75-04	Aluminum	Female	1.50 (38.1)	.88 (22.4)	1.13 (28.7)
27 8/	5.85 - 12.4	1.25:1	---	0.05	M85/ 4-008	M3922/ 75-04	Aluminum	Male	1.50 (38.1)	.88 (22.4)	1.13 (28.7)
28 8/	5.85 - 12.4	1.25:1	---	0.05	M85/ 4-008	M3922/ 75-04	Aluminum	Female	1.50 (38.1)	.88 (22.4)	1.13 (28.7)

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-A-22641).
5/ VSWR and RF insertion loss may be measured by swept frequency techniques.
6/ The internal dimensions and material of the waveguide shall be in accordance with MIL-W-85.
7/ Flange material and flange mounting hole dimensions shall be in accordance with MIL-F-3922.
8/ Reduced height waveguides and flanges.

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NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Navy - OS
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